Alessandro Simoni, Ph.D.

Deep Learning Research Engineer

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Links: alexj94.github.io | LinkedIn | GitHub | Google Scholar

Summary

I am a Deep Learning Research Engineer with 2+ years of experience in industry developing AI solutions for real-world industrial applications. My work has covered image reconstruction, diffusion models, semantic segmentation, 3D human pose estimation, and text-to-audio models. I earned a Ph.D. in Deep Learning and Computer Vision from University of Modena and Reggio Emilia in April 2024, focusing on 3D human-centric scene understanding and 3D object reconstruction. I am passionate about research and contributing to advancements in the field of Deep Learning.

Work Experience

Covision Lab April 2024 – (Now)
Deep Learning Research Engineer Brixen, Italy

- **Activities**: Diffusion Models, Text-to-Audio Models, Discrete Tomography, Semantic Segmentation, Classification, 3D Pose Estimation

AImageLab (University of Modena and Reggio Emilia) May 2020 – April 2024
Research Fellow / PhD Student Modena, Italy

- **Activities**: 3D Object Reconstruction (neural rendering, NeRF), 3D Skeleton Pose Estimation

Education

University of Modena and Reggio Emilia November 2020 – April 2024
Ph.D. in Deep Learning and Computer Vision Modena, Italy

- **Thesis**: *From Images to 3D Space: The Role of Semantic Keypoints for 3D Perception*

University of Modena and Reggio Emilia September 2017 – February 2020
MSc in Computer Science Modena, Italy

University of Modena and Reggio Emilia September 2013 – February 2017
BSc in Computer Science Modena, Italy

Skills

Spoken languages: Italian (mothertongue), English (C1 certified), Spanish (B1)
Programming languages: Python (proficiency), C/C++ (basic)
Libraries & Tools: PyTorch, Numpy, OpenCV, ONNX, TensorRT, VSCode
Deep Learning: 3D Pose Estimation, 3D Reconstruction, Image Diffusion Models, Text-to-Audio Models, Image Reconstruction, Semantic Segmentation, Image Classification

Relevant Publications (* = equal contribution, full list on Google Scholar)

SnapPose3D: Diffusion-Based Single-Frame 2D-to-3D Lifting of Human Poses 2026
A. Simoni, R. Catalini, D. Di Nucci, G. Borghi, D. Davoli, L. Garattoni, G. Francesca, Y. Kawana, R. Vezzani ICPR

Bounding Box-Guided Diffusion for Synthesizing Industrial Images and Segmentation Map 2025
A. Simoni *, F. Pelosin * CVPRW

3D Pose Nowcasting: Forecast the future to improve the present 2025
A. Simoni *, F. Marchetti *, G. Borghi, F. Becattini, L. Seidenari, R. Vezzani, A. Del Bimbo CVIU

Dynamic Label Injection for Imbalanced Industrial Defect Segmentation 2024
E. Caruso *, F. Pelosin *, **A. Simoni** *, M. Boschetti ECCVW

KRONC: Keypoint-based Robust Camera Optimization for 3D Car Reconstruction 2024
D. Di Nucci *, **A. Simoni** *, M. Tomei, L. Ciuffreda, R. Vezzani, R. Cucchiara ECCVW

Depth-Based Privileged Information for Boosting 3D Human Pose Estimation on RGB 2024
A. Simoni, F. Marchetti, G. Borghi, F. Becattini, D. Davoli, L. Garattoni, G. Francesca, L. Seidenari, R. Vezzani ECCVW

Depth-based 3D human pose pose refinement: Evaluating the refinet framework 2023
A. D'Eusanio, **A. Simoni**, S. Pini, G. Borghi, R. Vezzani, R. Cucchiara PRL

Semi-Perspective Decoupled Heatmaps for 3D Robot Pose Estimation from Depth Maps 2022
A. Simoni, S. Pini, G. Borghi, R. Vezzani IROS + RA-L

Multi-category mesh reconstruction from image collections 2021
A. Simoni *, S. Pini *, R. Vezzani, R. Cucchiara 3DV

Future urban scene generation through vehicle synthesis 2021
A. Simoni, L. Bergamini, A. Palazzi, S. Calderara, R. Cucchiara ICPR

Certifications

First Certificate in English (FCE) December 2020